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Finite Element Investigation of the Effects of Fiber Content and Orientation on the Mechanical Performance of Cotton Fiber-Reinforced Epoxy Composites

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Abstract: Cotton fiber-reinforced polymer composites have emerged as promising sustainable structural materials. However, the combined influence of fiber volume fraction (V_f) and fiber orientation on their mechanical performance has not been fully quantified. In this study, the mechanical behavior of cotton fiber-reinforced epoxy composites containing fiber volume fractions (V_f) of 20 vol.%, 40 vol.%, and 50 vol.% was systematically investigated through finite element analysis using ANSYS. Two representative reinforcement architectures, namely unidirectional and tri-directional fiber orientations, were modeled to evaluate their responses under tensile, flexural, and compressive loading conditions. The mechanical performance was assessed to elucidate the reinforcing mechanisms associated with different fiber configurations. It was found that the mechanical properties were consistently enhanced as the fiber volume fraction increased. Among all configurations, the composite containing 50 vol.% fiber volume fractions with a tri-directional orientation exhibited the highest mechanical performance, achieving tensile, flexural, and compressive strengths of 57.67, 71.03, and 68.31 MPa, respectively, corresponding to improvements of 46.1%, 17.5%, and 16.1% over the 20 vol.% tri-directional composites. Similarly, the 50 vol.% unidirectional composite demonstrated increases of 43.4% in tensile strength, 15.2% in flexural strength, and 17.3% in compressive strength relative to the corresponding 20 vol.% unidirectional configuration. Superior mechanical performance was consistently achieved by the tri-directional architecture compared with the unidirectional counterpart, which was attributed to a more uniform stress transfer mechanism, reduced stress concentration, and improved load redistribution, as confirmed by the von Mises stress distributions obtained from the finite element simulations. These findings demonstrate that both fiber volume fraction and fiber orientation play critical roles in governing the mechanical behavior of cotton fiber-reinforced epoxy composites. The numerical results further indicate that optimizing reinforcement architecture can substantially improve structural efficiency while maintaining the inherent sustainability and lightweight characteristics of natural fiber composites, thereby supporting their broader application in environmentally responsible engineering structures. 20%, 40%, and 50%.

Keywords: Cotton epoxy composite; Fiber content; Fiber orientation; Mechanical properties; Finite element analysis

1 Introduction

Recent years have witnessed a rapid surge in research on natural fiber composites due to their low cost, sustainability, and promising mechanical performance, especially for applications requiring eco-friendly alternatives to synthetic fibers [1]. Bio-composites, known as natural fiber-reinforced polymer composites, represent a growing

class of sustainable materials that merge plant-based fibers with polymer matrices. Fibers like jute, kenaf, hemp, and sisal are commonly used due to their renewability and environmental benefits. When combined with either biodegradable or traditional resins, these fibers help create materials that are not only lightweight and eco-friendly but also structurally reliable [2]. Natural fiber composites are considered environmentally friendly materials due to their biodegradable nature, which significantly reduces pollution. Additionally, these materials present a valuable opportunity to transform agricultural bio-waste into industrial products. These unique characteristics of natural fibers, including their sustainability and potential for waste utilization, are increasingly capturing the interest of researchers and engineers worldwide [3]. Cotton stands out as the most widely used natural fiber worldwide, accounting for a substantial portion of global fiber consumption [4]. Polymer matrix composites consist of a continuous polymer phase strengthened by high-performance fibers, which work together to improve the material's mechanical performance. The polymer matrix not only binds the fibers but also helps distribute applied loads and shields the reinforcement from environmental damage. Due to their impressive strength-to-weight ratio, these composites are increasingly seen as practical alternatives to metals, particularly in industries like aerospace, automotive, and construction. Their ease of processing using conventional techniques also adds to their appeal across a wide range of engineering applications [5]. The integration of natural fibers like cotton with polymer matrices is increasingly adopted in engineering applications such as automotive components, offering reduced weight, biodegradability, and mechanical properties comparable to synthetic fibers [6]. Finite element analysis allows simulation of stress–strain responses and failure modes in composite structures under various loadings [7].

Several researchers [8–10] have investigated the mechanical behavior of natural fiber composites under different loading conditions. Tasgin et al. [11] investigated the properties of epoxy composites reinforced with natural fibers (cotton, sisal, coir, and wool). The results showed that cotton achieved the highest tensile strength (52.81 MPa) and thermal conductivity (1.017 W/m·K), making it suitable for structural and heat dissipation applications. Sisal exhibited a high elastic modulus (1.69 GPa) and low friction coefficient (0.233), ideal for moving components. Coir fibers had the lowest thermal conductivity (0.187 W/m·K) but a high wear rate (4.615 mm/N·m), making them suitable for thermal insulation. Wool offered good flexibility with decent thermal insulation (0.518 W/m·K). These findings serve as a guide for selecting natural fibers based on application needs and highlight the potential for further development of eco-friendly composites. Wankhede et al. [12] studied cotton-epoxy composites (5–20 wt.%) and found that compressive strength peaked at 15 wt.% and impact strengths at 20 wt.%, though higher cotton content increased water absorption. Scanning electron microscopy analysis revealed fiber pull-out and matrix cracking as key failure modes, highlighting 15 wt.% as optimal for compressive performance and 20 wt.% for impact. Sagari and Prasad [13] investigated the effect of alkali treatment on woven jute and kenaf fiber-reinforced epoxy composites fabricated through hand lay-up and compression molding techniques. The mechanical performance was evaluated through tensile, flexural, and impact tests, while scanning electron microscopy was used to examine the fiber–matrix interface. The results demonstrated that alkali treatment significantly improved the mechanical properties and reduced void content by removing surface impurities, hemicellulose, and lignin from the fiber surface. Enhanced interfacial bonding between the fibers and epoxy matrix promoted more efficient load transfer, resulting in superior composite performance compared with untreated fiber composites. Mohamad Azani and Mohamad [14] aimed to enhance the utilization of natural fibers derived from agricultural waste in Malaysia (particularly bamboo and kenaf) as sustainable alternatives to traditional synthetic materials. Crack characteristics and crack tip opening angles of these fibers were simulated under quasi-static loading using ANSYS software, focusing on three weave patterns: 0°/90°, 30°/60°, and 45°/−45°. Balasubramanian et al. [15] focused on creating an eco-friendly and biodegradable hybrid composite with enhanced mechanical properties, utilizing natural fibers such as flax, aloe vera, and sisal, combined with epoxy resin. The composite was produced using the hand lay-up technique and underwent mechanical testing, including tension, flexural, impact, and hardness, according to the standards specified by the American Society for Testing and Materials (ASTM). Finite element analysis was performed using ANSYS 16.2, revealing a close match between experimental and analytical results, with deviations ranging from 2% to 6%.

Although existing studies have investigated the effect of fiber content and surface treatments on cotton fiber-reinforced epoxy composites, limited attention has been given to the role of fiber orientation, especially in more complex tri-directional configurations. Since fiber alignment significantly affects mechanical behavior under different loading conditions, this remains an area that warrants further exploration. This study contributes to the field by examining the combined effect of fiber volume fraction and fiber orientation on the mechanical performance of cotton/epoxy composites. Using finite element modeling, the research simulates tensile, compressive, and bending responses under both unidirectional and tri-directional fiber layouts. The insights gained aim to support the development of more efficient and sustainable composite material designs.

2 Research Methodology

2.1 Material Selection

In the present study, cotton fibers were used as reinforcement and epoxy resin as the matrix to produce composite samples with fiber volume fractions (V_f) of 20 vol.%, 40 vol.%, and 50 vol.%. This was done to investigate the influence of fiber content on the mechanical properties of the composites. The fibers were arranged in both unidirectional and tri-directional orientations to evaluate the effect of fiber alignment on the composite performance. The process began by defining the properties of the constituent materials, as summarized in Table 1. The properties of the epoxy matrix were obtained from the standard material database integrated within the ANSYS simulation environment, using Ansys Granta EduPack, Release 2024 R1 [16]. The selected epoxy material properties were subsequently used as input parameters for the representative volume element (RVE) analysis and subsequent composite laminate simulations. To ensure accurate material representation, baseline data for the structural epoxy resin were defined with a density of 1160 kg/m^3, tensile strength of 54.6 MPa, young's modulus of 3.78 GPa, Poisson's ratio of 0.35, bulk modulus of 4.2 GPa, and a shear modulus of 1.4 GPa. Regarding the cotton fiber reinforcement, baseline characteristics were adapted from comparative studies on natural cellulose fibers, incorporating the foundational framework provided by Taşdemir et al. [17] regarding sustainable bio-composite constituents. Because ANSYS Material Designer requires fixed, deterministic single-point values rather than experimental ranges to execute RVE homogenization, mean representative values derived from the literature ranges were implemented as simulation inputs.

Table 1. Mechanical properties of cotton fibers and epoxy resin

Property	Cotton Fiber	Epoxy Resin
Density (kg/m^3)	1550	1160
Tensile strength (MPa)	442	54.6
Young's modulus (GPa)	6.62	3.78
Poisson's ratio	–	0.35

Note: The symbol (–) designates parameters that are either non-applicable or not required as direct constitutive numerical inputs for that specific phase during homogenization and material characterization routines.

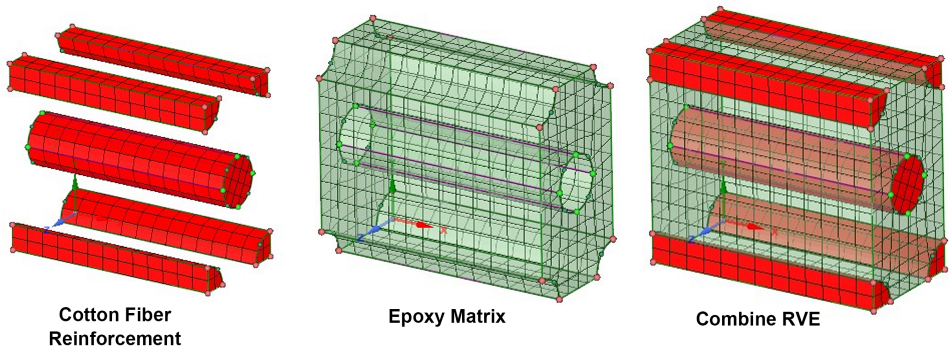


Figure 1. Meshed RVE model for 20% cotton fiber in epoxy

Note: RVE = Representative Volume Element.

Based on the above data, the effective mechanical properties of the cotton fiber-reinforced epoxy composites were determined using the Material Designer module in ANSYS Workbench. Since ANSYS Material Designer requires the fiber volume fraction (V_f) as input for representative volume element generation, the target fiber volume fractions of 20 vol.%, 40 vol.%, and 50 vol.% were achieved by prescribing corresponding fiber weight fractions (W_f) of 25.0%, 47.1%, and 57.2%, respectively. These weight fractions were converted using the densities of the cotton fibers (1550 kg/m^3) and epoxy resin (1160 kg/m^3) according to Eq. (1).

$$V_f = \frac{W_f/\rho_f}{(W_f/\rho_f) + (W_m/\rho_m)} \tag{1}$$

where, V_f is the fiber volume fraction, W_f and W_m are the weight fractions of the fiber and matrix, respectively, and ρ_f and ρ_m are the corresponding densities of the fiber and epoxy matrix. The calculated volume fractions were subsequently used to generate the RVE models. Periodic boundary conditions were applied to the RVE, and

numerical homogenization was performed to estimate the equivalent orthotropic elastic properties of the composite. Figure 1 illustrates the meshed RVE developed for the 20 vol.% fiber volume fractions, which represents the repeating microstructural unit used to predict macroscopic material behavior. The homogenized properties obtained from the RVE analysis were subsequently imported into ANSYS Composite PrepPost (ACP) for laminate modeling and mechanical simulations under tensile, flexural, and compressive loading conditions.

2.2 Mathematical Formulation

The approach is conceptually aligned with the classical Rule of Mixtures, which provides a first-order approximation for estimating the effective properties of composite materials [18, 19]. Under uniaxial loading conditions, and assuming perfect bonding between the fiber and matrix, both phases are subjected to the same strain (iso-strain condition), as defined in Eq. (2).

$$\varepsilon_f = \varepsilon_m = \varepsilon_c \tag{2}$$

where, ε_f is the strain in the fiber, ε_m is the strain in the matrix, and ε_c represents the overall strain in the composite. Assuming perfect bonding between fiber and matrix, the total stress in the composite can be approximated as a volume-weighted average of the individual phase stresses. When combined with Hooke's law, the effective young's modulus of the composite is described by the Voigt model, commonly used as an upper-bound estimate for composites with fibers aligned along the loading direction [20].

$$E_c = V_f \times E_f + V_m \times E_m \tag{3}$$

where, E_c is the effective young's modulus of the composite, V_f and V_m are the volume fractions of the fiber and matrix, respectively, E_f and E_m represent the young's moduli of the fiber and matrix.

When the fiber and matrix in a composite carry shear loads one after the other, as in unidirectional composites, the Reuss model (also known as the inverse Rule of Mixtures) is commonly used to estimate the in-plane shear modulus. This model assumes that both materials experience the same shear strain, leading to a harmonic average of their shear moduli, as shown in Eq. (4). It gives a lower-bound estimate for shear stiffness and is often used alongside the Voigt model, which provides an upper-bound estimate for stiffness in the fiber direction. Together, these models help verify the accuracy of simulated material properties [21].

$$\frac{1}{G_c} = \left(\frac{V_f}{G_f}\right) + \left(\frac{V_m}{G_m}\right) \tag{4}$$

where, G_c is the effective in-plane shear modulus of the composite, V_f and V_m are the volume fractions of the fiber and matrix, respectively, G_f and G_m represent the shear moduli of the fiber and matrix.

2.3 Computer-Aided Design Modeling

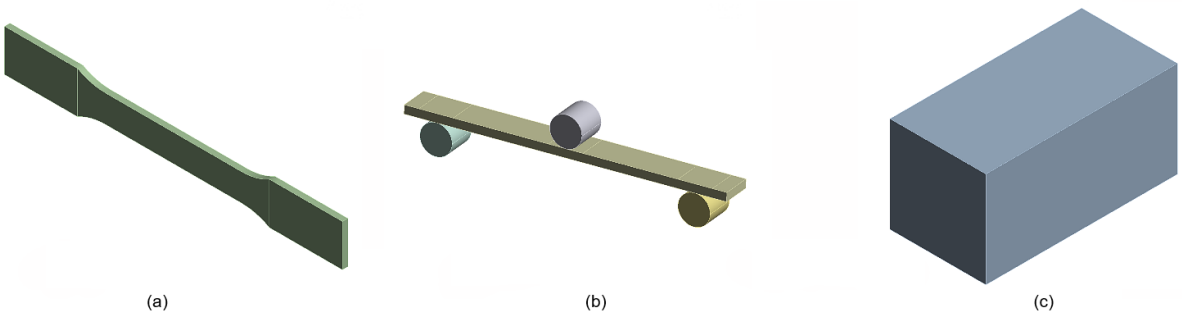


Figure 2. Isometric views of (a) tensile test specimen, (b) three-point bending specimen, (c) compression test specimen

The models were prepared using SpaceClaim. To investigate the mechanical response of epoxy composites reinforced with cotton fibers under tensile load, the computer-aided design of testing specimens was guided by ASTM D3039 [22]. Each tensile specimen had overall dimensions of 162 × 20 × 2.6 mm, as shown in Figure 2a. While ASTM D3039 standardly prescribes a constant rectangular cross-section coupon, the present numerical model incorporates enlarged end sections to facilitate the application of displacement constraints while minimizing boundary-induced stress concentrations, with the central gauge section retaining the required rectangular geometry. To assess the

flexural performance, simulations followed the three-point bending configuration outlined in ASTM D7264 [23]. Each specimen was modeled with dimensions of 100 × 10 × 2.7 mm and placed on a support span of 80 mm, as shown in Figure 2b. The compression analysis was developed with reference to ASTM D6641/D6641M; however, the numerical model did not explicitly reproduce the combined-loading compression fixture. A simplified displacement-controlled compression configuration was therefore employed to investigate the comparative compressive response of the different composite configurations, using specimens modeled with dimensions of 12.7 × 25.4 × 12.7 mm as shown in Figure 2c. ASTM D6641 [24].

2.4 Composite Modeling

The composite laminates were developed in ACP to investigate the influence of fiber orientation on the mechanical behavior of cotton fiber-reinforced epoxy composites. Two laminate configurations were considered for each fiber content (20%, 40%, and 50%): a unidirectional laminate and a tri-directional laminate. For the unidirectional configuration, all laminate plies were assigned a fiber orientation of 0°. This arrangement aligns the reinforcing fibers with the principal loading direction and serves as the reference configuration for evaluating the mechanical response. For the tri-directional configuration, the laminate consisted of alternating 0° and ±45° fiber orientations to distribute the applied load among multiple fiber directions. Table 2 illustrates the laminate configurations adopted for the unidirectional and tri-directional specimens used in the present study.

Table 2. Laminate configurations for the unidirectional and tri-directional specimens

Ply No.	Unidirectional Laminate	Tri-Directional Laminate
1	0°	0°
2	0°	+45°
3	0°	−45°
4	0°	0°
5	0°	+45°
6	0°	−45°
7	0°	0°
8	0°	+45°
9	0°	−45°

A local coordinate system (rosette) was defined in ACP to accurately control the fiber orientation of each ply. The rosette established the reference direction for the laminate, with the 0° direction aligned with the primary loading axis of the specimen. Individual ply orientations were subsequently assigned relative to this reference coordinate system, ensuring consistent fiber alignment throughout the laminate model. The same rosette definition and orientation strategy were applied to the tensile, flexural, and compression specimens to maintain consistency among the numerical models. The homogenized orthotropic properties obtained from the RVE analysis in ANSYS Material Designer were used as the effective material properties for the individual composite plies. For the evaluation of the tensile, flexural, and compressive responses, displacement-controlled loading was applied incrementally to the corresponding finite element models. The nonlinear material response was incorporated into the finite element simulations to represent the nonlinear stress–strain behavior observed in the composite specimens. The resulting ply-level stresses were evaluated using the Tsai–Wu failure criterion implemented in ACP. The Tsai–Wu failure index was monitored throughout the loading process, and failure initiation was defined when the failure index reached the critical value of FI = 1.

The Tsai–Wu criterion was used as the failure criterion rather than as the constitutive relation responsible for generating the nonlinear stress–strain response. Thus, the nonlinear material formulation governs the calculated material response under increasing displacement, while the Tsai–Wu criterion identifies the corresponding onset of composite failure. The reaction force corresponding to FI = 1 was taken as the limiting load for the respective loading configuration. The tensile strength was subsequently calculated from the limiting tensile load using [22]:

$$\sigma_t = \frac{P_f}{A} = \frac{P_f}{bt} \tag{5}$$

where, $P_f$is the reaction force at failure initiation, b is the specimen width, and t is the specimen thickness.

For the three-point bending configuration, the flexural strength was calculated as [23]:

$$\sigma_f = \frac{3P_f L}{2bt^2} \tag{6}$$

where, P_f is the failure-initiation load, L is the support span, b is the specimen width, and t is the specimen thickness.

The compressive strength was calculated from [24]:

$$\sigma_c = \frac{P_f}{A_c} \tag{7}$$

where, A_c is the effective cross-sectional area subjected to compression.

Accordingly, the reported tensile, flexural, and compressive strength values were determined from the load corresponding to the Tsai–Wu failure criterion rather than by directly interpreting maximum equivalent stress as the material strength. This distinction is important because the maximum equivalent stress was used only as a response parameter in the mesh-sensitivity assessment.

2.5 Mesh Sensitivity Analysis

To ensure the numerical reliability of the finite element model and assess the influence of mesh size on the predicted response, a mesh sensitivity analysis was conducted using the tensile model of the cotton fiber-reinforced epoxy composite containing 20% fiber content with unidirectional fiber orientation. Five global mesh sizes of 3.5, 3.0, 2.5, 2.0, and 1.5 mm were investigated. The maximum equivalent stress obtained from each mesh was used as the response parameter for comparison. The maximum equivalent stress increased from 37.355 MPa for the 3.5 mm mesh to 38.666 MPa for the 1.5 mm mesh. Taking the finest 1.5 mm mesh as the reference, the corresponding differences were 3.391%, 1.821%, 0.039%, and 0.0026% for the 3.5, 3.0, 2.5, and 2.0 mm meshes, respectively. The difference between the 2.5 mm and 1.5 mm meshes was only 0.039%, indicating that further refinement beyond 2.5 mm produced a negligible change in the predicted equivalent stress. Therefore, a 2.5 mm mesh was selected for the subsequent simulations as a suitable balance between numerical accuracy and computational efficiency. The mesh-convergence assessment was based primarily on the numerical results presented in Table 3, and Figure 3 illustrates the meshed composite laminate developed in ACP [25–27].

Table 3. Mesh independence study based on maximum equivalent stress

No.	Mesh Size (mm)	Number of Nodes	Number of Elements	Equivalent Stress (MPa)
1	3.5	338.0	287.0	37.355
2	3.0	464.0	396.0	37.962
3	2.5	623.0	538.0	38.651
4	2.0	954.0	835.0	38.665
5	1.5	1,612.0	1,491.0	38.666

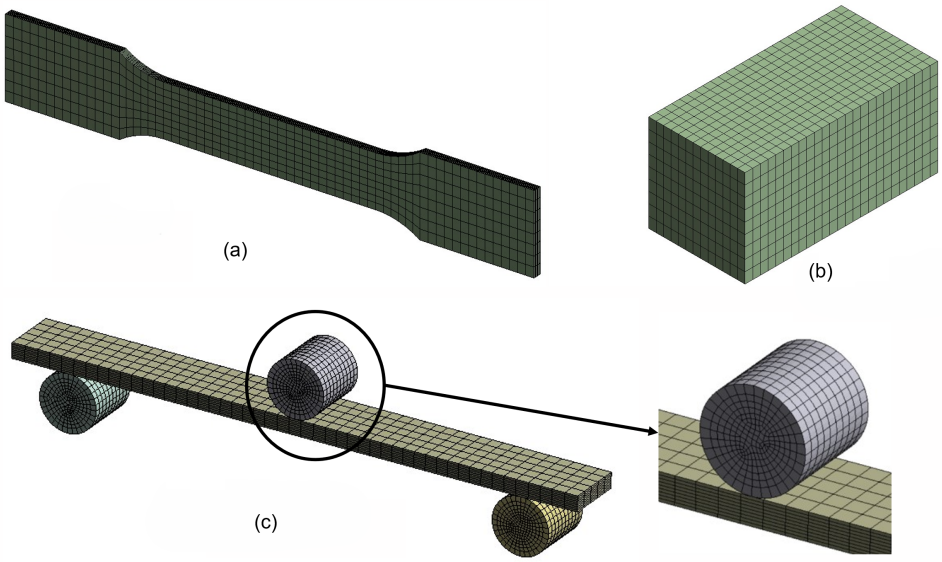


Figure 3. Meshed models of (a) tensile test specimen, (b) compression test specimen, and (c) three-point bending specimen

2.6 Interactions and Boundary Conditions

For present research, frictionless interactions were applied between all rollers and testing specimens for a three-point bending test, as shown in Figure 4. While the specimens used for tensile and compression tests are single parts, so no interactions were required for them.

For the tensile and compression simulations, a fixed support (right end in the case of the tensile specimen and bottom face in the case of compression) was applied as a boundary condition for the specimen. A tensile and compressive displacement was applied as a load on each specimen, as shown in Figure 5a and Figure 5b.

For the three-point bending simulation, fully fixed support was applied at the base of both lower rollers, while a ramped displacement load was applied vertically at the midpoint of the upper roller, as shown in Figure 5c.

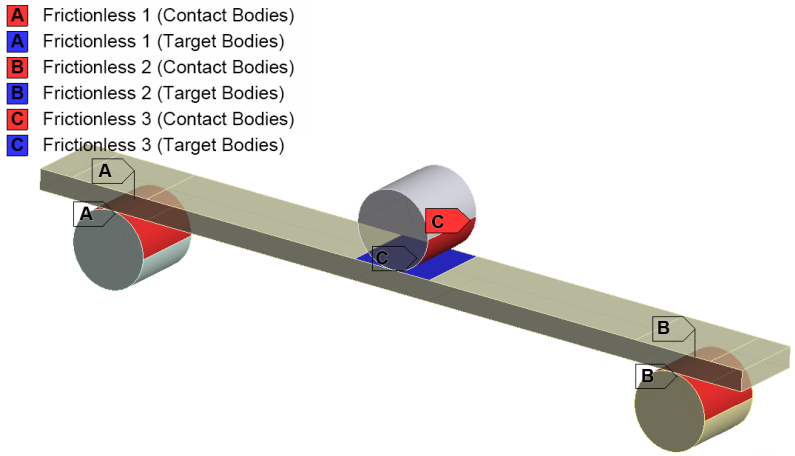


Figure 4. Frictionless contact definition between supports and beam for performing a three-point bending test

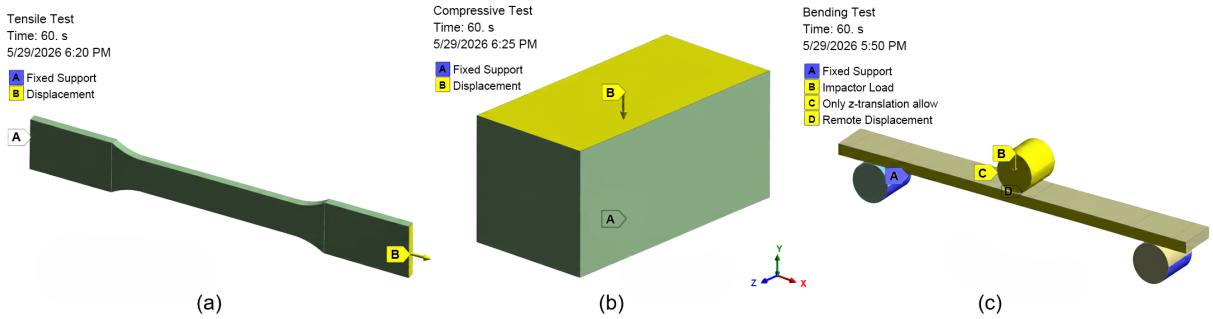


Figure 5. Load and boundary conditions used for (a) tensile test specimen, (b) compression test specimen, (c) three-point bending specimen

3 Results and Discussion

3.1 Validation of Results

The finite element model was validated by comparing the predicted mechanical properties with the experimental results reported by Soni et al. [25] in their study on cotton fiber-reinforced epoxy polymer composites. The validation considered both tensile and flexural strengths for neat epoxy (0% fiber) and composites containing 20% cotton fiber reinforcement. Figure 6 presents the comparison between the present numerical predictions and the experimental data available in the literature. For tensile loading, the finite element model predicted strengths of 36.596 MPa and 37.03 MPa for the 0% and 20% fiber composites, respectively, compared with the experimental values of 36.29 MPa and 37.04 MPa. The corresponding percentage differences were only 0.843% and 0.027%, indicating excellent agreement between the numerical and experimental results. Similarly, the flexural validation results presented in Figure 7 demonstrated close correspondence with the experimental observations. The predicted flexural strengths for the 0% and 20% fiber composites were 90.85 MPa and 59.592 MPa, while the corresponding experimental values were 91.23 MPa and 58.78 MPa, resulting in percentage differences of 0.417% and 1.381%, respectively.

The small deviations observed between the numerical and experimental results can be attributed to the idealized assumptions employed in the finite element model, including uniform fiber distribution, perfect fiber-matrix interfacial bonding, and the absence of manufacturing defects such as voids or fiber misalignment. In contrast, experimental composites are subject to material variability and fabrication-induced imperfections that can influence their mechanical response. Nevertheless, the maximum deviation remained below 1.5% for both tensile and flexural properties, demonstrating that the developed finite element model provides reliable predictions of the mechanical behavior of cotton fiber-reinforced epoxy composites and can be confidently employed for subsequent parametric investigations.

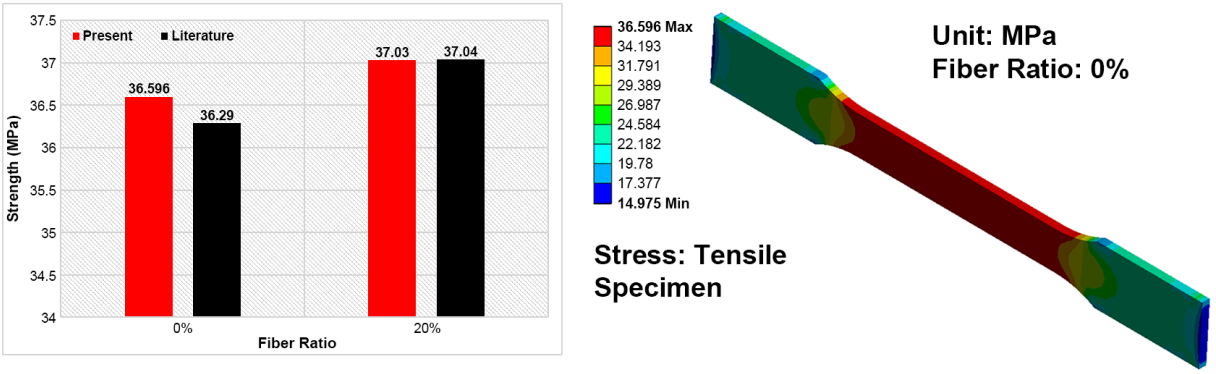


Figure 6. Validation of the proposed method against experimental tensile strength data reported by Soni et al. [25]

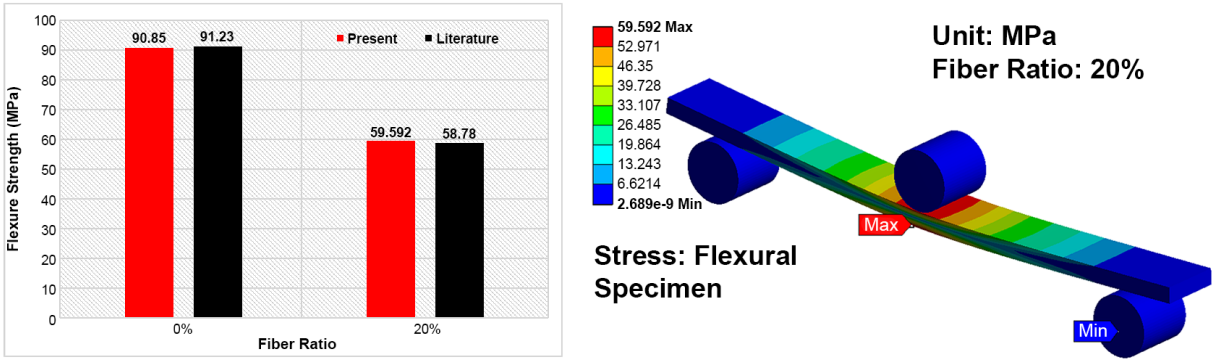


Figure 7. Validation of the proposed method against experimental flexural strength data reported by Soni et al. [25]

3.2 Tensile Strength Results

Tensile strength is a key property in determining a material's resistance to elongation and fracture when subjected to uniaxial tensile loads. In this study, tensile tests were conducted on epoxy composites reinforced with cotton fibers at different fiber volume fractions (20 vol.%, 40 vol.%, and 50 vol.%), using both unidirectional and tri-directional fiber orientations. The results are presented in Figure 8. A consistent improvement in tensile strength was observed with the increase in fiber content. At 20 vol.% fiber volume fractions, the tensile strength was 38.66 MPa for the unidirectional fiber orientation and 39.47 MPa for the tri-directional fiber orientation. As the fiber content increased to 40 vol.%, tensile strength rose to 48.95 MPa and 50.42 MPa, respectively. The highest values were obtained at 50 vol.%, with 55.46 MPa for the unidirectional fiber orientation and 57.68 MPa for the tri-directional fiber orientation. Compared with the 20 vol.% composites, the 50 vol.% composites exhibited increases in tensile strength of approximately 43.4% and 46.1% for the unidirectional and tri-directional configurations, respectively. Moreover, the tri-directional orientation consistently produced higher tensile strength than the unidirectional configuration across all fiber volume fractions, confirming the role of fiber orientation in enhancing mechanical behavior. The improved performance of tri-directional samples is attributed to more effective load distribution across different planes, reducing localized stress concentration and potential premature failure.

Although the stress–strain curves for both fiber orientations appear generally similar in shape, as shown in Figure 9, the tri-directional specimens consistently demonstrated higher peak stress values. This suggests that even though the overall deformation behavior is comparable, fiber orientation still has a significant impact on the material's

strength. By aligning fibers in multiple directions, the composite can distribute loads more effectively, which helps minimize stress concentrations and ultimately enhances tensile performance.

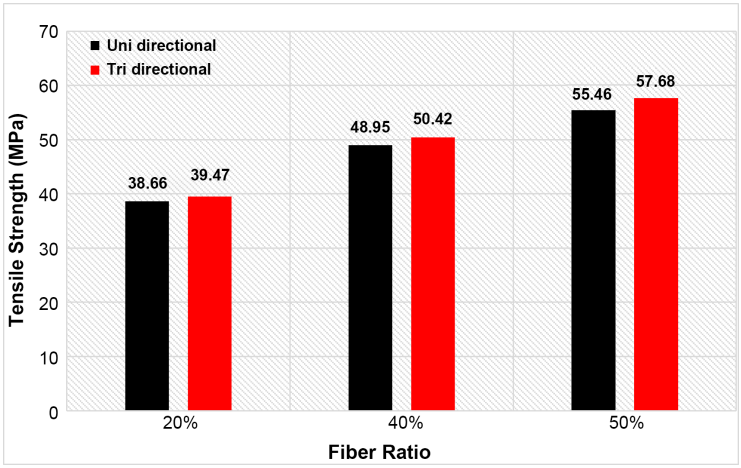


Figure 8. Tensile strength vs. fiber content for both orientations

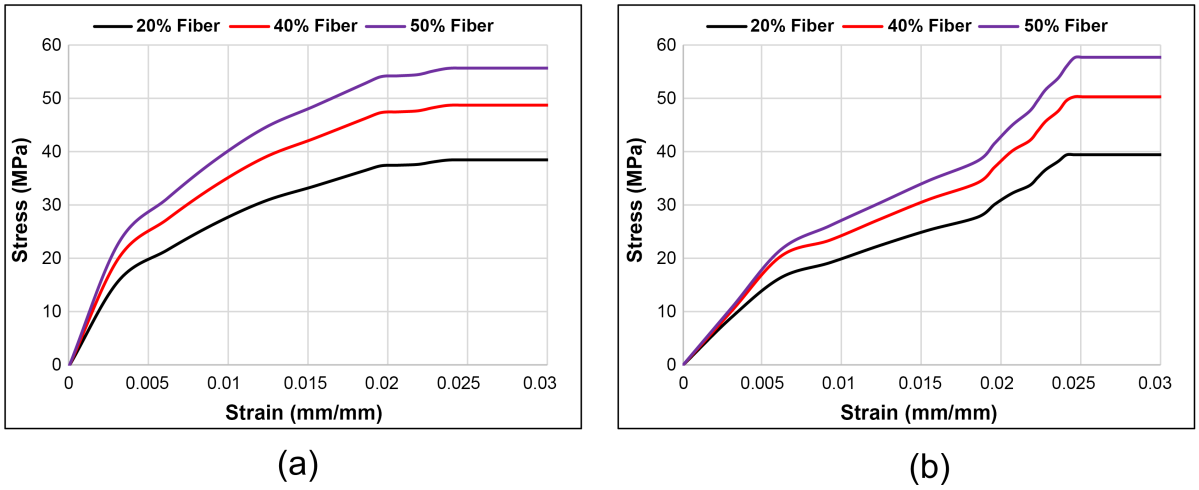


Figure 9. Tensile stress–strain curves for (a) unidirectional and (b) tri-directional fiber orientations

3.3 Flexure Strength Results

Flexural strength reflects a composite's resistance to bending forces and its ability to withstand deformation under flexural load. As illustrated in Figure 10, flexural strength increased with higher fiber content and showed a clear benefit from tri-directional orientation. At 20 vol.%, flexural strength was 59.86 MPa for the unidirectional fiber orientation and 60.46 MPa for the tri-directional fiber orientation. At 40 vol.%, values increased to 63.85 MPa for the unidirectional fiber orientation and 65.13 MPa for the tri-directional fiber orientation. The highest strength was recorded at 50 vol.%, reaching 68.96 MPa for the unidirectional fiber orientation and 71.03 MPa for the tri-directional fiber orientation. Compared with the 20 vol.%, composites, the 50 vol.%, composites exhibited relative improvements in flexural strength of approximately 15.2% and 17.5% for the unidirectional and tri-directional configurations, respectively. The results confirm that flexural performance is significantly influenced by both fiber quantity and orientation. Tri-directional orientation offers improved stiffness and stress distribution under bending loads, which is especially beneficial in structural applications subject to frequent flexural stress.

Under bending loads, the stress–strain response revealed that composites with tri-directional fiber orientation demonstrated greater resistance to flexural stresses compared to their unidirectional counterparts, as shown in Figure 11. This improved performance is largely due to the ability of multi-directional fiber alignment to withstand the complex stress distribution that occurs during bending, particularly the simultaneous tensile and compressive forces acting on opposite sides of the specimen. By distributing fibers across multiple planes, the composite can more effectively dissipate internal stresses and delay the onset of failure. This structural advantage not only increases

the peak flexural strength but also enhances the material's energy absorption capacity and overall durability under flexural loading.

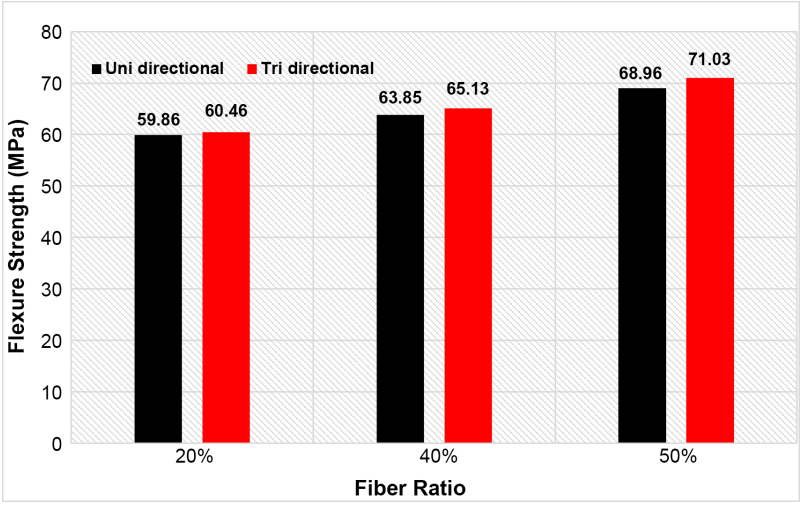


Figure 10. Flexural strength vs. fiber content for both orientations

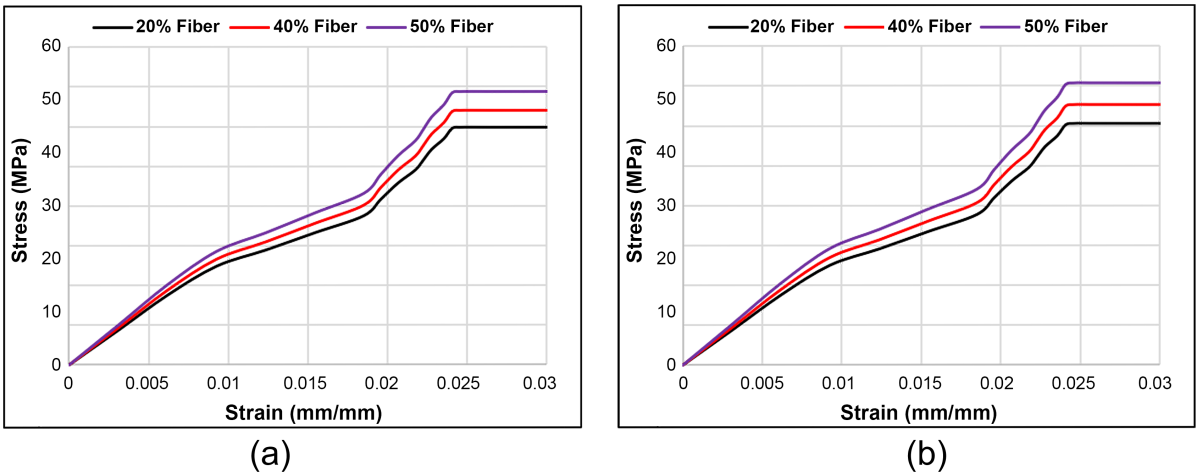


Figure 11. Flexural stress–strain curves for (a) unidirectional and (b) tri-directional fiber orientations

3.4 Compressive Strength

Compressive strength is an important parameter for evaluating the ability of a composite material to withstand compressive loading without failure. The compressive strength results are presented in Figure 12. At 20 vol.%, the calculated compressive strengths were 57.66 MPa for the unidirectional configuration and 58.81 MPa for the tri-directional configuration. Increasing the fiber content to 40 vol.%, resulted in compressive strengths of 64.69 MPa and 66.31 MPa for the unidirectional and tri-directional configurations, respectively. At 50 vol.%, fiber content, the maximum compressive strengths were obtained, with values of 67.64 MPa for the unidirectional configuration and 68.31 MPa for the tri-directional configuration. Relative to the 20 vol.%, fiber composites, the 50 vol.%, fiber composites showed increases in compressive strength of approximately 17.3% and 16.1% for the unidirectional and tri-directional configurations, respectively.

The compressive stress–strain responses shown in Figure 13 exhibit nonlinear behavior for both fiber orientations over the investigated fiber-content range. The stress initially increases with increasing strain and subsequently approaches a plateau near the maximum load level. The tri-directional configuration consistently produced slightly higher compressive strength values than the corresponding unidirectional configuration at all investigated fiber contents. This difference can be attributed to the different fiber orientations and their influence on the distribution and transfer of compressive loads within the composite. In the unidirectional configuration, the fibers are aligned primarily along the principal loading direction, whereas the ±45° plies in the tri-directional configuration provide additional

load-transfer directions. Consequently, the different laminate architectures produce distinct stress distributions under compression.

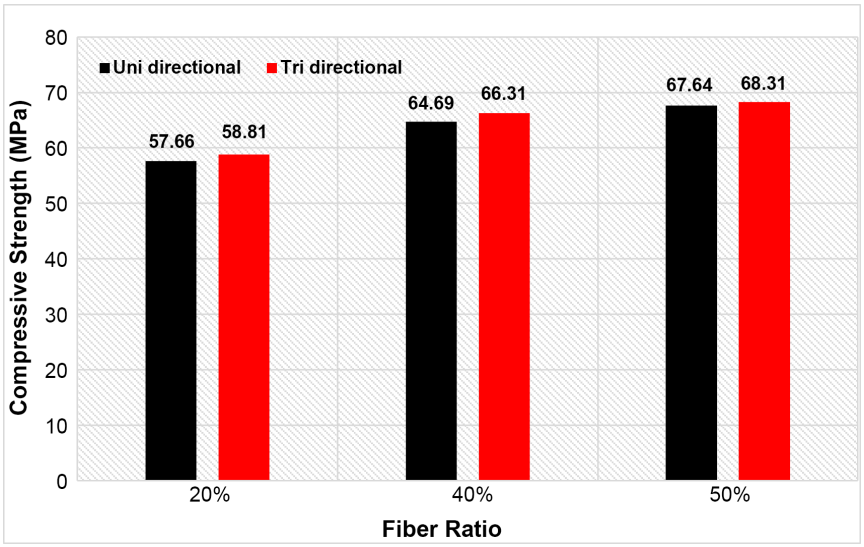


Figure 12. Compressive strength vs. fiber content for both orientations

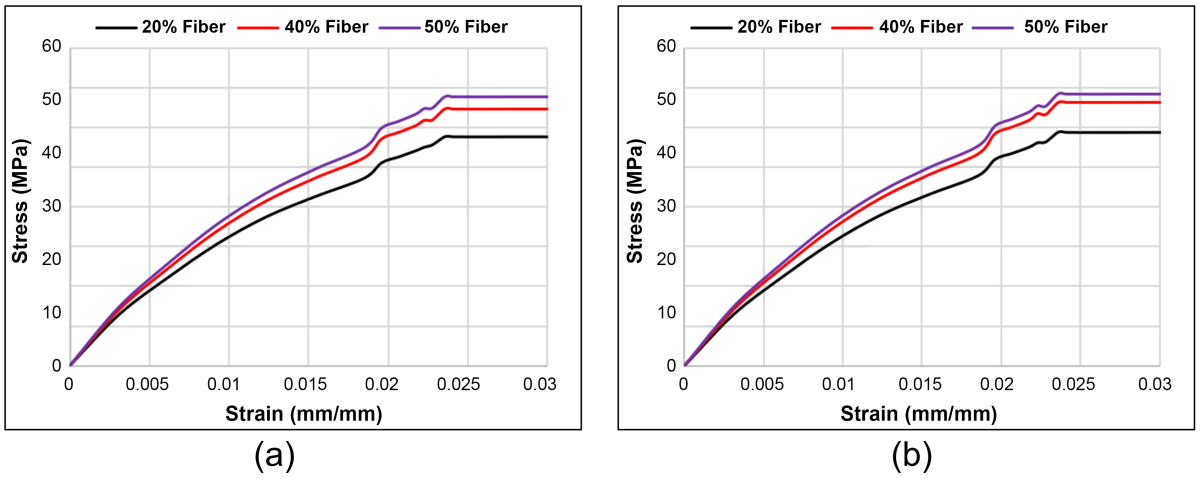


Figure 13. Compression stress–strain curves for (a) unidirectional and (b) tri-directional fiber orientations

The results indicate that increasing fiber content generally increases the calculated compressive strength for both laminate configurations. The present finite element analysis evaluates the overall compressive response and failure initiation using the adopted material formulation and Tsai–Wu failure criterion. Therefore, specific damage mechanisms such as fiber micro buckling, matrix crushing, fiber fracture, or delamination are not directly quantified in the present analysis. The observed differences in compressive response should consequently be interpreted in terms of the influence of fiber content and laminate orientation on the calculated structural response and failure initiation.

3.5 Numerical Visualization of Stress

For the unidirectional laminate (Figure 14a), the stress contour shows a relatively localized stress distribution, with a peak von Mises stress of 55.457 MPa concentrated in the central gauge section along the 0° fiber direction. The stress difference across the specimen cross-section is approximately 32.56 MPa, with a minimum value of 22.897 MPa. This distribution indicates that the 0° plies carry a substantial portion of the applied tensile load because the fiber direction is aligned with the principal loading direction. The concentration of stress in the central gauge region identifies this area as the critical region of the tensile model. However, the von Mises stress contour alone does not establish a specific physical failure mechanism. The onset of composite failure was therefore evaluated

using the Tsai–Wu failure criterion at the ply level. For the tri-directional laminate (Figure 14b), the peak von Mises stress is 57.676 MPa, with a minimum value of 23.662 MPa. The stress distribution differs from that of the unidirectional laminate because the 0° and ±45° plies provide different material orientations for carrying the applied load. The off-axis plies contribute to the redistribution of stresses among the laminate layers, resulting in a different stress distribution compared with the unidirectional configuration. The calculated stress distribution demonstrates the influence of fiber orientation on load transfer within the laminate. Any conclusion regarding failure initiation is based on the Tsai–Wu failure index rather than on the von Mises contour alone. Therefore, the observed stress distribution should be interpreted as a numerical representation of load transfer and stress concentration rather than direct evidence of matrix cracking, fiber fracture, or delamination.

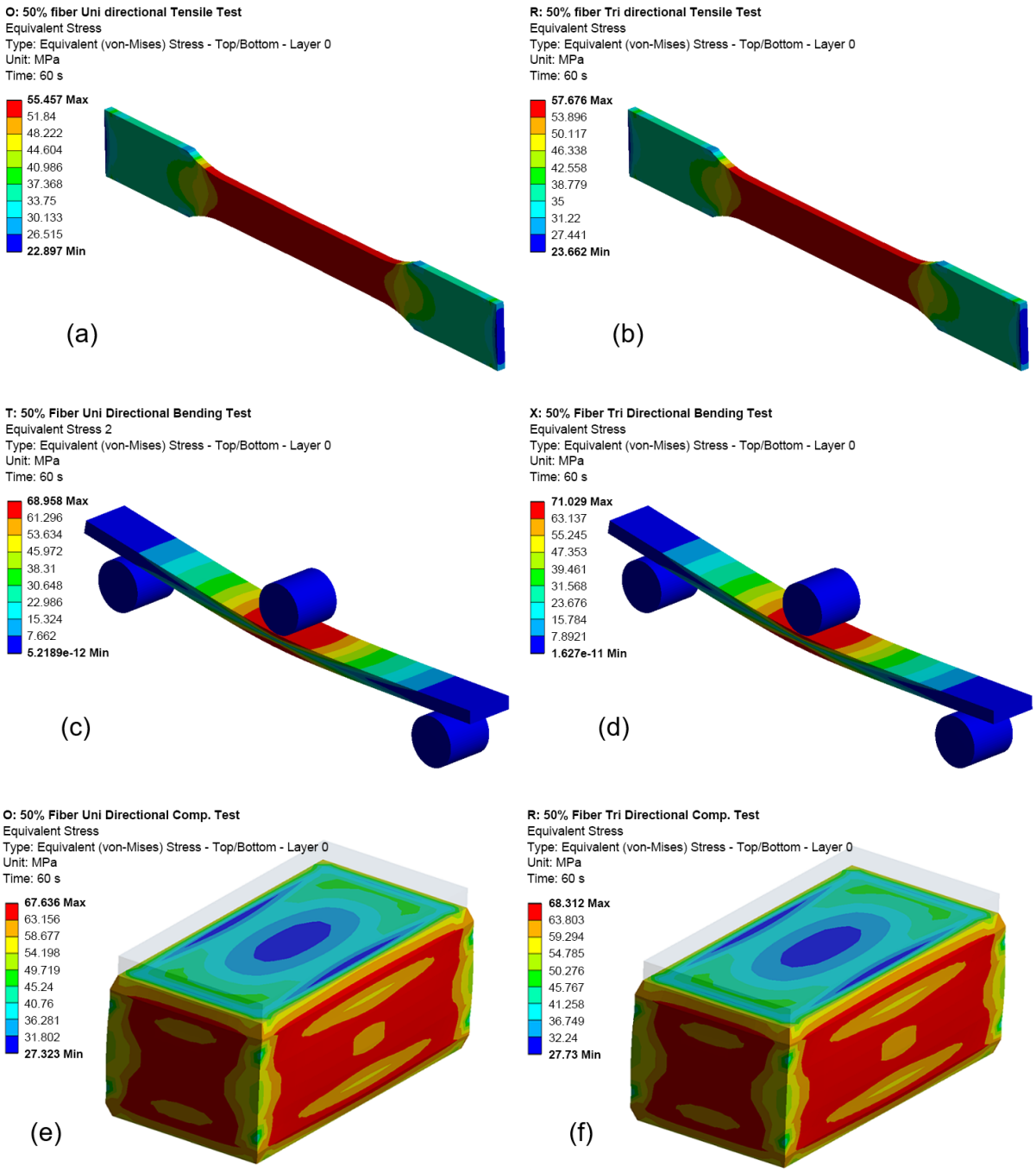


Figure 14. Von Mises stress distribution for the 50% fiber content composite under different loading conditions: (a) tensile unidirectional; (b) tensile tri-directional; (c) flexural unidirectional; (d) flexural tri-directional; (e) compressive unidirectional; (f) compressive tri-directional

Under three-point bending, the unidirectional laminate (Figure 14c) exhibits a pronounced stress gradient through the specimen thickness, with a peak von Mises stress of 68.958 MPa occurring near the bottom surface of the specimen. The stress decreases toward the neutral axis, where the bending stress approaches a lower value. This distribution is consistent with the variation of bending stress across the laminate thickness, with the region farther from the neutral axis experiencing a greater stress level. Since the 0° fibers are aligned with the principal material direction, the corresponding plies contribute substantially to the load-carrying response. Nevertheless, the stress contour does not by itself demonstrate matrix cracking, fiber breakage, or delamination. The failure state was instead assessed using the Tsai–Wu failure index. The tri-directional laminate (Figure 14d) exhibits a peak von Mises stress of 71.029 MPa. The presence of +45° and −45° plies change the distribution of stresses among the laminate layers and provides additional material orientations for carrying the bending-induced stresses. The difference between the stress distributions of the unidirectional and tri-directional configurations demonstrates the influence of laminate orientation on the structural response under flexural loading. However, specific damage mechanisms such as crack propagation or delamination cannot be directly identified from the von Mises stress contours used in the present analysis. Consequently, the discussion is limited to the calculated stress distribution and the corresponding Tsai–Wu failure response.

For the unidirectional laminate under compression (Figure 14e), the stress contour shows a relatively localized distribution in the central region, with a peak von Mises stress of 67.636 MPa and a minimum value of 27.323 MPa toward the outer regions. The concentration of stress in the central region indicates that this area experiences a relatively high level of compressive loading. Because the 0° fibers are aligned with the primary loading direction, the laminate response is strongly influenced by the properties of the material in this direction. The tri-directional laminate (Figure 14f) exhibits a more distributed stress pattern, with the 0° and ±45° plies contributing to the overall load transfer. The different fiber orientations modify the stress distribution through the laminate thickness and provide additional directions for carrying the applied compressive load. The calculated stress contour therefore demonstrates the effect of laminate architecture on the compressive response. However, the present analysis does not explicitly resolve fiber microbuckling, kink-band formation, or progressive compressive damage. Such failure mechanisms would require an appropriate mode-specific failure and progressive-damage formulation. Therefore, the present results are interpreted using the calculated stress distribution and Tsai–Wu failure index rather than assigning the observed stress concentrations to a particular physical failure mode.

Overall, the results demonstrate that fiber orientation has a significant influence on the stress distribution of the cotton fiber-reinforced epoxy laminates under tensile, flexural, and compressive loading. The 0° plies primarily contribute to load transfer along the principal loading direction, while the +45° and −45° plies modify the distribution of stresses among the laminate layers. The von Mises stress contours identify regions of relatively high and low stress, whereas the Tsai–Wu failure criterion provides the basis for identifying the onset of composite failure. Since no mode-specific progressive-damage model was implemented, the present analysis does not directly quantify matrix cracking, fiber fracture, delamination, micro buckling, fatigue damage accumulation, or progressive failure. These mechanisms are therefore not used as direct interpretations of the numerical contours.

3.6 Comparison With Previous Studies

The results of the present study were compared with three recent investigations on cotton fiber-reinforced epoxy composites, as summarized in Table 4. Wankhede et al. [12] investigated randomly distributed cotton fibers and reported only compressive properties, with strengths ranging from 50 to 55 MPa. Owen et al. [26] combined experimental testing with finite element analysis for chemically treated woven cotton fabric composites and reported a maximum tensile strength of 43.7 MPa. Pawar and Joshi [27] experimentally investigated epoxy composites reinforced with layered cotton fibers and cashew nut shell liquid-modified epoxy, achieving tensile and flexural strengths of approximately 48 MPa and 80 MPa, respectively. The present finite element study investigated the influence of fiber content and laminate orientation using unidirectional and tri-directional configurations at fiber volume fractions of 20 vol.%, 40 vol.%, and 50 vol.%. Among the investigated configurations, the 50 vol.% tri-directional laminate exhibited the highest predicted tensile, flexural, and compressive strengths of 57.7 MPa, 71.0 MPa, and 68.3 MPa, respectively. Compared with the findings of Owen et al. [26], the predicted tensile strength was approximately 32% higher (57.7 MPa versus 43.7 MPa), while it exceeded the tensile strength reported by Pawar and Joshi [27] by approximately 20% (57.7 MPa versus 48 MPa). Although the predicted flexural strength was lower than the 80 MPa reported by Pawar and Joshi [27], their study employed a cashew nut shell liquid-modified epoxy matrix and layered reinforcement architecture, which differ from the material system and laminate configuration considered in the present work. Overall, the results demonstrate that increasing the cotton fiber content and adopting a tri-directional laminate configuration contribute to more efficient stress distribution and enhanced mechanical performance. The numerical predictions showed excellent agreement with the published experimental validation data, with maximum deviations below 1.5%, confirming the reliability of the proposed finite element modeling approach.

Table 4. Comparison of the present study with recent investigations on cotton fiber-reinforced epoxy composites

Study	Tensile Strength (MPa)	Flexural Strength (MPa)	Compressive Strength (MPa)
Wankhede et al. [12]	–	–	50–55
Owen et al. [26]	43.7	38.6	–
Pawar and Joshi [27]	48.0	80.0	–
Present study	57.7	71.0	68.3

Note: "–" indicates that the property is not applicable or the data are not provided.

4 Conclusion

The present study numerically investigated the tensile, flexural, and compressive behavior of cotton fiber-reinforced epoxy composites using finite element analysis. The effects of fiber volume fraction (V_f of 20 vol.%, 40 vol.%, and 50 vol.%) and laminate configuration (unidirectional and tri-directional) were systematically evaluated to examine their influence on the calculated mechanical response. The results showed that increasing the fiber content generally enhanced the predicted mechanical strength of the composites. Relative to the 20 vol.% composites, increasing the fiber content to 50 vol.% resulted in increases in tensile strength of 43.4% and 46.1%, flexural strength of 15.2% and 17.5%, and compressive strength of 17.3% and 16.1% for the unidirectional and tri-directional configurations, respectively. The tri-directional configurations generally exhibited slightly higher strength values than the corresponding unidirectional configurations, particularly at higher fiber contents. This behavior indicates that laminate orientation influences the distribution and transfer of applied loads within the composite. Among the investigated configurations, the 50 vol.% tri-directional composite exhibited the highest calculated tensile, flexural, and compressive strength values. However, these results should be interpreted within the scope of the adopted finite element model and investigated loading conditions rather than as a general indication of superior performance for all applications. The developed finite element model showed good agreement with the available published experimental results, with deviations below 1.5% for the considered validation cases. This agreement supports the applicability of the adopted numerical framework for evaluating the influence of fiber content and laminate configuration on the mechanical response of cotton fiber-reinforced epoxy composites. The Tsai–Wu failure criterion was used to evaluate failure initiation within the composite, while the present analysis does not explicitly resolve individual damage mechanisms such as matrix cracking, fiber fracture, micro buckling, or delamination.

Overall, the findings demonstrate that fiber volume fraction and laminate orientation are important parameters governing the calculated mechanical response of cotton fiber-reinforced epoxy composites. The finite element framework provides a computational approach for investigating these parameters and can support preliminary material and laminate design studies, potentially reducing the number of experimental configurations required during the early stages of composite development. Further experimental investigation and failure-mode-specific modeling would be beneficial for validating the predicted behavior and for examining the detailed damage mechanisms of the composites.

Author Contributions

Conceptualization, A.Y., M.N.I., and M.M.; methodology, A.Y., M.N.I., and M.M.; formal analysis, M.S., A.Y., and U.U.; validation, M.M. and N.U.; data curation, M.S., A.Y., and U.U.; writing—original draft preparation, A.Y., M.N.I., M.S., M.M., N.U., and U.U.; writing—review and editing, A.Y., M.N.I., M.S., M.M., N.U., and U.U. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation of this manuscript, AI-assisted language tools were used solely for language improvement, grammar correction, and readability enhancement. The authors reviewed and verified all content and take full responsibility for the accuracy and integrity of the manuscript.

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